ClipSAT : Cambridge & Edexcel Additional Mathematics 0606/4PM1

Unit 4.01: Logarithmic & Exponential Functions

Name: _____

Q1. Solve the exponential equation 2^{x+2} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{2ln 2}{ln 5 - ln 2}
[ ] B: x = frac{2ln 5}{ln 5 - ln 2}
[ ] C: x = frac{ln 2}{2ln 5 - ln 2}
[ ] D: x = frac{2}{ln 5 - ln 2}

Q2. Solve the exponential equation 2^{x+3} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{3ln 2}{ln 5 - ln 2}
[ ] B: x = frac{3ln 5}{ln 5 - ln 2}
[ ] C: x = frac{3}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{3ln 5 - ln 2}

Q3. Solve the exponential equation 2^{x+4} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{4ln 2}{ln 5 - ln 2}
[ ] B: x = frac{ln 2}{4ln 5 - ln 2}
[ ] C: x = frac{4ln 5}{ln 5 - ln 2}
[ ] D: x = frac{4}{ln 5 - ln 2}

Q4. Solve the exponential equation 2^{x+5} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{5ln 2}{ln 5 - ln 2}
[ ] B: x = frac{5ln 5}{ln 5 - ln 2}
[ ] C: x = frac{5}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{5ln 5 - ln 2}

Q5. Solve the exponential equation 2^{x+6} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{6ln 5}{ln 5 - ln 2}
[ ] B: x = frac{6ln 2}{ln 5 - ln 2}
[ ] C: x = frac{6}{ln 5 - ln 2}
[ ] D: x = frac{ln 2}{6ln 5 - ln 2}

Q6. Solve the exponential equation 2^{x+7} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{7ln 2}{ln 5 - ln 2}
[ ] B: x = frac{ln 2}{7ln 5 - ln 2}
[ ] C: x = frac{7ln 5}{ln 5 - ln 2}
[ ] D: x = frac{7}{ln 5 - ln 2}

Q7. Solve the exponential equation 2^{x+8} = 5^x , giving x in terms of common logarithms.

[ ] A: x = frac{8ln 2}{ln 5 - ln 2}
[ ] B: x = frac{8ln 5}{ln 5 - ln 2}
[ ] C: x = frac{ln 2}{8ln 5 - ln 2}
[ ] D: x = frac{8}{ln 5 - ln 2}

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Q8. Solve the simultaneous system: log_2x - log_2y = 2 and x - 3y = 3 . What is the value of x ?

[ ] A: 12

[ ] B: 3

[ ] C: 6

[ ] D: 14